Microgrid Economic MPC — Plant + System Modeling Design

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Abstract

A *discrete-time linear time-invariant (LTI) state-space plant model* for a grid-tied microgrid with PV, BESS, and load, designed as the input to a downstream Economic MPC (EMPC) formulation. The state is the battery state-of-charge (SoC, scalar). Controls are grid import and battery power. Disturbances are PV power, load demand, and electricity price (perfect forecasts in v1). Time scale is 1 hour, prediction horizon 24 hours. The objective function (formulated by the MPC-formulation teammate) is a linear-in-control stage cost $c(k)\, u_{\text{grid}}(k)\, \Delta T$ with a quadratic terminal cost on SoC deviation and a tiny Tikhonov regularizer — this makes the downstream optimization a convex QP solvable with MATLAB `quadprog` in milliseconds. All hand-off code is documented in a README contract so the MPC-formulation, simulation, and report teammates can integrate without further negotiation.

Deliverable owner: Kholifah Zein Nuril Musthafa (Group 7).
Deadline: 2026-06-10 23:59 WIB (5 days from now).
Status: Design — awaiting team approval.

Contents

1 System Architecture

The modeled microgrid is a 4-component system on a common AC bus, grid-tied:

$$\begin{array}{rcl} P_{\text{pv}}(k) & \longrightarrow & \\ & & \text{AC Bus} \\ u_{\text{batt}}(k) & \longrightarrow & \text{(Modeled)} \\ & & \longrightarrow u_{\text{grid}}(k) \\ P_{\text{load}}(k) & \longleftarrow & \end{array}$$

where $P_{\text{load}}(k)$ is a forecasted exogenous disturbance, $P_{\text{pv}}(k)$ is a forecasted PV output, $u_{\text{batt}}(k)$ is the controlled battery power, and $u_{\text{grid}}(k)$ is the controlled grid import.

- **PV array**: forecasted, curtailable (v1: no explicit curtailment; see Section 10).
- **BESS (battery)**: 1 dynamic state (SoC); 1 control input (charge/discharge power).
- **Load**: forecasted, exogenous.
- **Grid**: controlled import. Export is not allowed in v1 (flagged as v2).

Sign convention (matches Cortes-Aguirre et al. 2024, arXiv:2412.10851):

- $u_{\text{batt}} > 0 =$ **discharging** (battery delivers to AC bus; SoC decreases).
- $u_{\text{batt}} < 0 =$ **charging** (battery absorbs from AC bus; SoC increases).
- $u_{\text{grid}} > 0 =$ **importing** from grid.
- $u_{\text{grid}} < 0 =$ **exporting** to grid (NOT allowed in v1; lower bound = 0).
- $P_{\text{pv}} \geq 0$, $P_{\text{load}} \geq 0$.

2 State-Space Plant Model

This is the central deliverable. The matrices (A, B, C, D, E, F_e) are passed to the MPC-formulation teammate.

2.1 Variables and dimensions

Symbol	Meaning	Dim.	Unit
$x(k)$	Battery state-of-charge (SoC), normalized	$\mathbb{R}$	dimensionless
$u(k) = [u_{\text{grid}}(k), u_{\text{batt}}(k)]^\mathsf{T}$	Control inputs	$\mathbb{R}^2$	kW
$d(k) = [P_{\text{pv}}(k), P_{\text{load}}(k), c(k)]^\mathsf{T}$	Disturbances (forecast, perfect in v1)	$\mathbb{R}^3$	kW, kW, \$/kWh
$y(k) = [u_{\text{grid}}(k), x(k)]^\mathsf{T}$	Outputs	$\mathbb{R}^2$	kW, dim.
ΔT	Sampling time	scalar	1 hour

2.2 State equation

$$x(k+1) = A\,x(k) + B\,u(k) + E\,d(k) \tag{1}$$

with matrices

$$A = \begin{bmatrix}1\end{bmatrix}, \qquad B = \begin{bmatrix}0 & -\dfrac{\Delta T}{E_{\text{nom}}}\end{bmatrix}, \qquad E = \begin{bmatrix}0 & 0 & 0\end{bmatrix}. \tag{2}$$

Derivation of B: In one hour, discharging at $u_{\text{batt}} > 0$ removes energy $u_{\text{batt}} \cdot \Delta T$ from the battery. Normalized: $\Delta x = -(u_{\text{batt}} \cdot \Delta T)/E_{\text{nom}}$, so

$$x(k+1) = x(k) + \Delta x = x(k) - \frac{u_{\text{batt}}(k) \cdot \Delta T}{E_{\text{nom}}}.$$

Grid import u_{grid} does not directly affect SoC. Disturbances do not directly affect SoC in v1.

Efficiency simplification (v1): Round-trip efficiency is 100% in the dynamics. The penalty for losses appears only in the cost (Section 4) to keep dynamics linear and the QP convex. (This is Cortes-Aguirre 2024's deliberate choice, §II-D, justified as "preserves convexity.") Asymmetric $\eta_c \neq \eta_d$ is flagged for v2.

2.3 Output equation

$$y(k) = C\,x(k) + D\,u(k) + F_e\,d(k) \tag{3}$$

with

$$C = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \qquad F_e = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{4}$$

Derivation of D and F_e (output 1 = grid import): From power balance,

$$u_{\text{grid}}(k) = P_{\text{load}}(k) - P_{\text{pv}}(k) - u_{\text{batt}}(k),$$

so output 1 is $y_1 = u_{\text{grid}} = -u_{\text{batt}} + P_{\text{load}} - P_{\text{pv}}$. Output 2 is $y_2 = x$, the SoC.

2.4 Power-balance equality constraint

Embedded in the output equation but also listed as a QP equality constraint for clarity:

$$P_{\text{load}}(k) - P_{\text{pv}}(k) - u_{\text{batt}}(k) - u_{\text{grid}}(k) = 0 \quad \forall k \in \{0, \ldots, N_p - 1\}. \tag{5}$$

This eliminates u_{grid} from the free decision variables if the MPC-formulation teammate prefers; the choice is theirs. **Recommended**: keep both u_{grid} and u_{batt} as free controls and add this as an equality, so the cost $c(k)\,u_{\text{grid}}(k)\,\Delta T$ directly captures the electricity bill.

3 Prediction Matrices Φ and Γ (assignment Section 1.2 item 3)

The standard stacked MPC prediction form (Rawlings, Mayne, Diehl 2017, Chapter 2). **This is the form the assignment explicitly asks for** ("membentuk matriks prediksi F dan Φ"). None of the 4 papers surveyed use this form (Cortes-Aguirre uses CVX directly, Polanco Vásquez uses algebraic constraints, Vasilj uses MILP, Jia uses ADMM iterations) — but the form is equivalent, and the assignment requires it.

3.1 Stacked vectors

$$\begin{aligned}
X &= \left[x(k+1|k),\, x(k+2|k),\, \ldots,\, x(k+N_p|k)\right]^{\mathsf{T}} \in \mathbb{R}^{N_p} \\
U &= \left[u(k|k),\, u(k+1|k),\, \ldots,\, u(k+N_p-1|k)\right]^{\mathsf{T}} \in \mathbb{R}^{2N_p} \\
D &= \left[d(k|k),\, d(k+1|k),\, \ldots,\, d(k+N_p-1|k)\right]^{\mathsf{T}} \in \mathbb{R}^{3N_p}
\end{aligned}$$

where U interleaves $[u_{\text{grid}}, u_{\text{batt}}]$ per step and D interleaves $[P_{\text{pv}}, P_{\text{load}}, c]$ per step.

3.2 Stacked prediction equation

$$X = \Phi_x \cdot x(k) + \Gamma_x \cdot U + \Psi_x \cdot D \tag{6}$$

For our model ($A = 1$, $B = [0, -\Delta T/E_{\text{nom}}]$, $E = 0$):

- $\Phi_x = [1, 1, \ldots, 1]^{\mathsf{T}} \in \mathbb{R}^{N_p}$ (column of N_p ones — every future state starts from the current one with gain 1).

- Γ_x is lower-triangular block-Toeplitz. With $b = [0, -\Delta T/E_{\text{nom}}]$:

$$\Gamma_x = \begin{bmatrix} b & 0 & 0 & \cdots & 0 \\ 2b & b & 0 & \cdots & 0 \\ 3b & 2b & b & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ N_p b & (N_p-1)b & (N_p-2)b & \cdots & b \end{bmatrix} \in \mathbb{R}^{N_p \times 2N_p}. \tag{7}$$

- $\Psi_x = 0$ (disturbances do not affect state in v1).

3.3 Code to construct Γ_x

Provided to the MPC-formulation teammate as `build_prediction_matrices.m`:

```
function [Phi_x, Gamma_x, Psi_x] = build_prediction_matrices(Np, dT, E_nom)
    % Construct the stacked prediction matrices for the microgrid plant.
    %
    % Inputs:
    %   Np    - prediction horizon (e.g. 24)
    %   dT    - sampling time in hours (e.g. 1.0)
    %   E_nom - BESS nominal energy in kWh (e.g. 5.0)
    %
    % Outputs:
    %   Phi_x   - Np x 1   (state contribution)
    %   Gamma_x - Np x 2Np (control contribution)
    %   Psi_x   - Np x 3Np (disturbance contribution, zero in v1)

    Phi_x = ones(Np, 1);
    b = [0, -dT / E_nom];          % 1x2 row of B
    Gamma_x = zeros(Np, 2*Np);
    for i = 1:Np
        for j = 1:i
            Gamma_x(i, 2*(j-1)+1 : 2*j) = (i - j + 1) * b;
        end
    end
    Psi_x = zeros(Np, 3*Np);
end
```

Listing 1: build_prediction_matrices.m

4 Economic Objective

The MPC-formulation teammate constructs the QP cost $J = \frac{1}{2}U^\top HU + f^\top U$ from three ingredients.

4.1 Stage cost (linear in control — drives "economic" behavior)

$$L(k) = c(k) \cdot u_{\text{grid}}(k) \cdot \Delta T. \tag{8}$$

Interpretation: at hour k, importing u_{grid} kW at price $c(k)$ (in \$/kWh) yields a bill of $c(k) \cdot u_{\text{grid}}(k) \cdot$ 1h. Multiply by ΔT to get kWh.

4.2 Terminal cost (quadratic — keeps SoC reachable at horizon end)

$$V_f(x(N_p)) = \lambda \cdot (x(N_p) - x_{\text{target}})^2. \tag{9}$$

$x_{\text{target}} = 0.5$ (return battery to half charge by end of horizon). $\lambda > 0$ is a tuning parameter (recommend $\lambda = 1.0$ for v1; sweep $\pm 10\times$ in analysis).

Origin: This is the single-layer simplification of Vasilj et al. 2019's two-layer "cost-to-go V_s" and Cortes-Aguirre et al. 2024's terminal cost V_f (their Eq. 12, more complex). Our simple quadratic keeps the QP well-posed.

4.3 Tikhonov regularization (quadratic — makes the QP well-conditioned)

$$R(u) = \varepsilon \cdot \|u(k)\|^2, \tag{10}$$

$\varepsilon \approx 10^{-3}$ (very small). Physical meaning: penalize sudden swings in battery/grid power to keep the closed-loop trajectory smooth.

4.4 Total cost

$$J = \sum_{k=0}^{N_p-1} \left[c(k) \cdot u_{\text{grid}}(k) \cdot \Delta T + \varepsilon \cdot \|u(k)\|^2 \right] + \lambda \cdot (x(N_p) - x_{\text{target}})^2. \tag{11}$$

4.5 QP matrices (H, f) the teammate builds

After substituting $X = \Phi_x x(k) + \Gamma_x U$:

$$\begin{aligned} H_q &= 2\lambda \Gamma_x^\mathsf{T} \Gamma_x, & f_q &= 2\lambda(\Phi_x x(k) - x_{\text{target}})^\mathsf{T} \Gamma_x, \\ H_r &= 2\varepsilon I_{2N_p}, & f_r &= 0, \\ H_l &= 0, & f_l &= c_{\text{full}}, \end{aligned}$$

where $c_{\text{full}} = [c(0), 0, c(1), 0, \ldots, c(N_p - 1), 0]^\mathsf{T} \cdot \Delta T \in \mathbb{R}^{2N_p}$.

Total: $H = H_q + H_r$ (must be positive definite; ε ensures this), $f = f_q + f_l$.

This is a **convex QP** that MATLAB's `quadprog(H, f, A_ineq, b_ineq, A_eq, b_eq, lb, ub)` solves in milliseconds.

4.6 v2 extensions (out of v1 scope, mentioned for context)

- **Peak shaving:** add $\alpha \cdot \max(0, u_{\text{grid}}(k) - P_{\text{peak}})^2$ per hour. Needs slack or epigraph reformulation to keep QP.
- **Renewable utilization:** add $\beta \cdot P_{\text{pv,curtailed}}(k)^2$ if v1 is extended to allow PV curtailment.
- **Battery degradation:** add $\gamma \cdot |u_{\text{batt}}(k)|$ (cycle-cost) or $\gamma \cdot u_{\text{batt}}(k)^2$ (loss proxy).
- **Multi-objective (Polanco Vásquez 2023):** add $\delta \cdot \mathrm{CO_2}(P_{\text{grid}}, P_{\text{DG}}, \ldots)$.
- **Time-varying λ, ε:** could be made horizon-dependent.

5 Constraints

All in QP form. The MPC-formulation teammate builds the matrices; we just specify them.

#	Type	Constraint	Mathematical form
1	Equality	Power balance	$u_{\text{grid}} - u_{\text{batt}} = P_{\text{load}} - P_{\text{pv}}$
2	Inequality	SoC lower bound	$x(k) \geq \text{SoC}_{\text{min}}$ for $k = 1, \ldots, N_p$
3	Inequality	SoC upper bound	$x(k) \leq \text{SoC}_{\text{max}}$ for $k = 1, \ldots, N_p$
4	Box	Battery power	$-P_{\text{batt,max}} \leq u_{\text{batt}}(k) \leq P_{\text{batt,max}}$
5	Box	Grid import	$0 \leq u_{\text{grid}}(k) \leq P_{\text{grid,max}}$ (no export in v1)
6	Inequality (optional)	Terminal SoC	$x(N_p) \geq \text{SoC}_{\text{min,terminal}}$ (e.g. 0.3)

The SoC constraints (#2, #3) are linear inequalities on U after substitution $X = \Phi_x x(k) + \Gamma_x U$. Specifically:

$$\begin{aligned} \text{Lower:} \quad & \Gamma_x U \geq \text{SoC}_{\text{min}} \cdot \mathbf{1} - \Phi_x x(k) \implies -\Gamma_x U \leq \Phi_x x(k) - \text{SoC}_{\text{min}} \cdot \mathbf{1}, \\ \text{Upper:} \quad & \Gamma_x U \leq \text{SoC}_{\text{max}} \cdot \mathbf{1} - \Phi_x x(k). \end{aligned}$$

6 Parameters (placeholder values; to be confirmed by team)

Parameter	Symbol	Value	Unit	Notes
BESS nominal energy	E_{nom}	5.0	kWh	Lab-scale lithium. Replace with team spec.
BESS max power	$P_{\text{batt,max}}$	2.5	kW	0.5C rate
Grid max import	$P_{\text{grid,max}}$	10.0	kW	UI lab grid connection
PV installed	$P_{\text{pv,installed}}$	3.0	kW_p	UI rooftop estimate
Load peak	$P_{\text{load,peak}}$	4.0	kW	UI building scale
SoC min	SoC_{min}	0.2	–	Lithium longevity
SoC max	SoC_{max}	0.8	–	Lithium longevity
Initial SoC	$x(0)$	0.5	–	Midpoint
Sampling time	ΔT	1.0	hour	Matches assignment plot granularity
Prediction horizon	N_p	24	hours	Matches assignment plot length
Target SoC at horizon end	x_{target}	0.5	–	Midpoint
Terminal cost weight	λ	1.0	\$/kWh^2 (equiv.)	Tune in analysis
Tikhonov weight	ε	10^{-3}	–	Small for conditioning

Action item: The team should confirm or replace these with the actual data available. If real measurements exist from a lab microgrid, use them. If not, these are the values to use in the report and to generate synthetic disturbances (see Section 7).

7 Forecast Data (disturbances)

Provided as a single function `generate_forecasts.m` that returns a 24×3 matrix $[P_{\text{pv}} \mid P_{\text{load}} \mid c]$.

7.1 PV forecast $P_{\mathbf{pv}}(k)$

Synthetic: clipped sinusoid peaking at solar noon.

$$P_{\text{pv}}(k) = \begin{cases} P_{\text{pv,installed}} \cdot \max\big(0, \sin(\pi(k-6)/12)\big), & k = 6, \dots, 18 \\ 0, & \text{otherwise.} \end{cases}$$

Real-data alternative: NREL TMY data for Jakarta (

$$-6.2^\circ$$

lat), interpolated to hourly.

7.2 Load forecast $P_{\mathbf{load}}(k)$

Synthetic: typical residential/commercial profile with two peaks (morning + evening).

$$\begin{aligned} P_{\text{load}}(k) = 0.5 \cdot P_{\text{load,peak}} & \qquad \text{(overnight, } k = 0\text{–}5,\ 22\text{–}23) \\ + 0.8 \cdot P_{\text{load,peak}} + 0.2 P_{\text{load,peak}} \sin(\cdot) & \qquad \text{(morning peak, } k = 6\text{–}9) \\ + 0.4 \cdot P_{\text{load,peak}} & \qquad \text{(midday, } k = 10\text{–}16) \\ + 0.9 \cdot P_{\text{load,peak}} + 0.1 P_{\text{load,peak}} \sin(\cdot) & \qquad \text{(evening peak, } k = 17\text{–}21) \end{aligned}$$

Real-data alternative: actual UI campus load profile if accessible.

7.3 Electricity price $c(k)$

TOU (time-of-use) tariff. Two options:

- **Indonesian PLN "WBP/LWBP" categories** (WBP = peak 18:00–22:00, LWBP = off-peak): $c_{\text{WBP}} = 1.5 \times c_{\text{base}}$, $c_{\text{LWBP}} = c_{\text{base}}$. (Specific values per PLN regulation; team to confirm.)
- **Synthetic (for reproducibility):** $c(k) = c_{\text{base}}$ for $k = 0$–17, 22–23; $c(k) = 2.5 \times c_{\text{base}}$ for $k = 18$–21.

c_{base} is a placeholder; team to confirm with current PLN rate. Suggest $c_{\text{base}} = 1500\,\text{IDR/kWh} \approx$ \$0.10/kWh as a starting reference (matches Cortes-Aguirre 2024's \$0.10/kWh).

8 Simulation (not in your piece; document the contract)

The simulation teammate runs closed-loop over 7 days (or 24 h for the assignment plots):

```
for k = 0, 1, 2, ...
    % 1. Measure current SoC
    x_k = x_history(end);

    % 2. Build QP
    [H, f] = build_qp(x_k, forecast_k, params, lambda, epsilon);
    [Aeq, beq, Aineq, bineq, lb, ub] = ...
        build_constraints(Np, params, x_k, forecast_k);

    % 3. Solve QP
    U_star = quadprog(H, f, Aineq, bineq, Aeq, beq, lb, ub);
```

```
    % 4. Apply first control
    u_k = U_star(1:2);

    % 5. Advance dynamics
    x_next = A*x_k + B*u_k + E*d_k;

    % 6. Record for plotting
    x_history = [x_history, x_next];
    u_history = [u_history, u_k];
    c_history = [c_history, c_k];
end
```

Listing 2: Closed-loop simulation pseudocode

Required plots (per assignment example image):

1. Electricity price $c(k)$ over 24 h
2. Battery power $u_{\text{batt}}(k)$ over 24 h (positive = discharge)
3. SoC trajectory $x(k)$ over 24 h or 7 days
4. Total cost $\sum c(k)\, u_{\text{grid}}(k)\, \Delta T$ cumulative over days
5. Grid import $u_{\text{grid}}(k)$ over 24 h

Baseline for comparison: "No MPC" — always import $P_{\text{load}} - P_{\text{pv}}$ directly, ignore battery. Quantify cost reduction (should be non-trivial per the literature; e.g. Cortes-Aguirre 2024 got 2% annual cost reduction with a more sophisticated model).

9 File Hand-off Package (in /program/)

This is what you commit to `Group7_MicrogridMPC/program/`. Each file is documented in a top-of-file comment block.

File	Purpose	Inputs	Outputs
`plant_model.m`	Returns A, B, C, D, E, F_e matrices	$(E_{\text{nom}}, \Delta T)$	A, B, C, D, E, F_e
`build_prediction_matrices.m`	Constructs Φ_x, Γ_x, Ψ_x	$(N_p, \Delta T, E_{\text{nom}})$	Φ_x, Γ_x, Ψ_x
`build_constraints.m`	Constructs $A_{\text{eq}}, b_{\text{eq}}, A_{\text{ineq}}, b_{\text{ineq}}, lb, ub$	$(N_p, \text{params}, x(k), \text{forecasts})$	constraint matrices
`build_qp.m`	Given $(x(k), \text{forecast})$, returns H, f	$(x(k), \text{forecast}, \text{params})$	(H, f)
`parameters.m`	Returns all numerical parameters	none	struct with all Section 7 values
`generate_forecasts.m`	Returns 24×3 disturbance matrix	$(\text{date}, P_{\text{pv,installed}}, P_{\text{load,peak}}, c_{\text{base}})$	$D = [P_{\text{pv}}, P_{\text{load}}, c]$
`README.md`	Interface contract for teammates	none	documentation

`README.md` explicitly states:

- Variable names and units (kW, kWh, \$/kWh, h, dimensionless).
- Sign convention (grid import positive, battery discharge positive).
- v1 limitations (perfect forecast, no PV curtailment, no export).
- v2 extensions roadmap.
- MATLAB version tested.
- Required toolboxes (Optimization Toolbox for `quadprog`; otherwise `qpOASES` or `OSQP` via YALMIP).

10 Risks, Edge Cases, and Open Items

10.1 Known risks and mitigations

- **Infeasible QP** (e.g. load exceeds PV + battery + grid max): add slack variables on SoC bounds with penalty $M \cdot \varepsilon_{\text{slack}}$ (large M). Cortes-Aguirre 2024 uses hard bounds and accepts occasional infeasibility; v1 can match this and log a warning.
- **Solver failure** (numerical issue with `quadprog`): return previous u, log warning, continue simulation. Document the failure rate in the report.
- **Forecast error** (v1 assumes perfect): explicitly stated in the report's "Limitations" section. v2: add ±10–20% error to forecasts, show that the closed-loop cost is robust (Polanco Vásquez 2023 showed robust to 12% error).
- **PV overproduction** (when $P_{\text{pv}} > P_{\text{load}} + P_{\text{charge,max}}$): with v1's no-export and no-curtailment, this is infeasible. Two v2 options: (a) allow export, (b) add $P_{\text{curtailed}}$ as a slack with zero cost.
- **Battery SoC hits limit** at the end of horizon: handled by the QP inequality constraints automatically.
- **Sign confusion** between the two D's (state-space D matrix vs. disturbance stacked vector): README explicitly disambiguates with different names (D_{ss} for state-space, D_{stacked} for disturbance vector).

10.2 Open items for the team to confirm

- **Real data vs synthetic:** does the team have access to actual UI lab microgrid data? If yes, replace synthetic forecasts.
- **Parameter values:** Section 7 values are placeholders. Confirm with team.
- **PLN tariff specifics:** exact $c_{\text{WBP}}/c_{\text{LWBP}}$ ratio and c_{base}. Look up current regulation.
- **Folder name:** the assignment says `/project_name/` with subdirs `/program /reports /presentation /references /images /data`. Group's project folder name to be decided (e.g. `Group7_MicrogridMPC`).
- **Reference style:** IEEE numbered [1], [2], ... or author-year? Indonesian academic norms suggest IEEE. Confirm with lecturer's preference.

10.3 What I am NOT doing (and where to find it in the literature)

- **Stochastic MPC** (scenario-based): see Polanco Vásquez 2023 (they test 12% error) and Vasilj 2019 (ARMA+Copula scenarios). Out of v1 scope; "Bonus Challenge" item in assignment.
- **Distributed EMPC** (multiple networked microgrids with ADMM): see Jia 2026. Out of v1 scope; "Bonus Challenge" item.
- **Robust MPC** (worst-case disturbance): not in any of the 4 papers surveyed; available in Rawlings/Mayne/Diehl textbook. Out of v1 scope.
- **Nonlinear battery** (voltage curve, SoC-dependent efficiency): out of v1 scope; "Open Items" in Cortes-Aguirre 2024.
- **Kalman filter for state estimation:** v1 assumes $x(k)$ is measured perfectly. v2: add Luenberger or Kalman filter on x. "Bonus Challenge" item in assignment.

11 Intellectual Lineage (for the report's "References" section)

To make the academic attribution explicit:

Design element	Primary source	Borrowed from	Sec.
1-state SoC, 2-control form	Cortes-Aguirre 2024 (arXiv:2412.10851)	–	§2
Power-balance equality	Cortes-Aguirre 2024 (Eq. 6c)	–	§2.4
100% efficiency in dynamics, loss in cost	Cortes-Aguirre 2024 (§II-D)	–	§2.2
Linear stage cost $c(k)u_{\text{grid}}\Delta T$	Cortes-Aguirre 2024 (Eq. 1)	Ellis 2014, Rawlings 2012	§4.1
Terminal cost $\lambda(x(N_p)-x_{\text{target}})^2$	Vasilj 2019 (V_s) and Cortes-Aguirre 2024 (V_f, Eq. 12)	–	§4.2
Tikhonov regularization $\varepsilon\|u\|^2$	Rawlings/Mayne/Diehl 2017, §2.6	–	§4.3
Prediction matrices Φ, Γ in stacked form	Rawlings/Mayne/Diehl 2017, Ch. 2	–	§3
Hourly ΔT, 24h horizon, perfect forecast	Cortes-Aguirre 2024 (coarsened from 15 min to 1 h)	–	§7, 8
Lab-scale parameters	**Not from any paper** — own estimates, flagged as placeholders	–	§7
Two-layer day-ahead + real-time (v2)	Vasilj 2019	–	–
Multi-objective cost + CO_2 (v2)	Polanco Vásquez 2023 (Pareto GA)	–	§4.6
Distributed ADMM (v2)	Jia 2026	–	–
Foundational EMPC definitions	Ellis, Durand, Christofides 2014 (663 citations)	Rawlings, Angeli, Bates 2012	–
Standard MPC textbook	Rawlings, Mayne, Diehl 2017	–	–

12 Acceptance Criteria

How you'll know the modeling piece is done:

- ☐ `plant_model.m` returns the matrices in §2.2 and §2.3 with a sanity check (determinant, dimensions).
- ☐ `build_prediction_matrices.m` returns Φ_x, Γ_x, Ψ_x as specified in §3.
- ☐ `build_constraints.m` returns the constraint matrices in §5.
- ☐ `build_qp.m` returns a positive-definite H matrix and the correct f vector.
- ☐ `generate_forecasts.m` returns a 24×3 matrix with non-negative entries.
- ☐ `parameters.m` returns a struct with all §7 values.
- ☐ `README.md` documents sign convention, units, and v1 limitations.
- ☐ A 5-line sanity-check script runs end-to-end: load parameters, generate forecast, build QP for a sample $x(0)$, solve, print optimal $u(0)$. The output should be a finite real vector.
- ☐ Closed-loop simulation (run by simulation teammate) converges over 7 days without QP infeasibility.
- ☐ All 7 references cited in Section 15 are in the report's reference list.

13 Schedule (5 days remaining)

Day	Date	Task	Deliverable
1	Fri 2026-06-05 AM	Design approved, files scaffolded	Spec committed
1	Fri 2026-06-05 PM	Implement `parameters.m`, `plant_model.m`, `generate_forecasts.m`	3 .m files + README skeleton
2	Sat 2026-06-06	Implement `build_prediction_matrices.m`, `build_constraints.m`, `build_qp.m`. Sanity-check script.	4 .m files + sanity-check passing
2	Sat 2026-06-06 PM	Hand off to MPC-formulation teammate; integration test	Joint working QP
3	Sun 2026-06-07	Industrial background write-up (Section 1 of report structure)	2–3 pages draft
3	Sun 2026-06-07 PM	System description with parameters table (Section 2 of report structure)	1–2 pages draft
4	Mon 2026-06-08	State-space derivation write-up (Pemodelan Sistem, 20%)	2–3 pages with equations
4	Mon 2026-06-08 PM	Final report integration; presentation slides draft	Report draft v1, slides v1
5	Tue 2026-06-09	Rehearsal, final review, edge case fixes	Report v2, slides v2
5	Tue 2026-06-09 PM	Submit via EMAS before 23:59 WIB	Submitted

Buffer day Wednesday 2026-06-10 is **deadline day**; submit by 18:00 to avoid last-minute issues.

14 Open Questions for the Team

1. **Reference style:** IEEE numbered or author-year? (Affects report format, not this spec.)
2. **Project folder name:** `Group7_MicrogridMPC` or something else?
3. **Real data vs synthetic:** does the team have access to real UI microgrid data?
4. **PLN tariff values:** what are the current WBP/LWBP rates?
5. **MATLAB version:** is the Optimization Toolbox available? If not, fallback is YALMIP + OSQP (free, but slower to set up).
6. **Commit cadence:** push to a shared Git repo (GitHub/EMAS), or share via EMAS file upload?

15 References

References

[1] C. Cortes-Aguirre, Y.-A. Chen, A. Ghosh, J. Kleissl, and A. Khurram, "Economic MPC with an Online Reference Trajectory for Battery Scheduling Considering Demand Charge Management," *arXiv preprint arXiv:2412.10851*, Dec. 2024, submitted to *IEEE Trans. Smart Grid*.

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[3] L. O. Polanco Vásquez, J. López Redondo, J. D. Álvarez Hervás, V. M. Ramírez, and J. L. Torres, "Balancing CO_2 emissions and economic cost in a microgrid through an energy management system using MPC and multi-objective optimization," *Applied Energy*, vol. 347, art. 120998, Sept. 2023, doi: 10.1016/j.apenergy.2023.120998.

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[5] M. Ellis, H. Durand, and P. D. Christofides, "A tutorial review of economic model predictive control methods," *J. Process Control*, vol. 24, no. 8, pp. 1156–1178, Aug. 2014, doi: 10.1016/j.jprocont.2014.03.010.

[6] J. B. Rawlings, D. Angeli, and C. N. Bates, "Fundamentals of economic model predictive control," in *Proc. 51st IEEE Conf. Decision and Control (CDC)*, Maui, HI, USA, 2012, pp. 3851–3861.

[7] J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. Nob Hill Publishing, 2017. [Online]. Available: https://www.nobhillpublishing.com/

End of spec. Awaiting user approval before writing the GSD planning artifacts (`PROJECT.md`, `REQUIREMENTS.md`, `ROADMAP.md`, `STATE.md`, `config.json`) into `/home/wawabobo/MATLAB/SKPA FINPRO/.planning/`